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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Alamgir Banu
Student Name	A.Shashank Reddy
Roll Number	2520090139
Branch	CSE-IT
Course Outcome	CO-6

Case Study Topic: Bangalore-to-Mumbai Logistics – 0/1 Knapsack for Cargo Loading

Work Done Summary:

In this case study, we developed a Java-based cargo loading optimization system using the **0/1 Knapsack Dynamic Programming algorithm**. A logistics operator must load a truck with a maximum carrying capacity of **24 tons**. Eight cargo consignments are available, each having a weight and payment value.

The objective is to select the combination of consignments that maximizes the total payment without exceeding the truck capacity. Since each consignment can either be fully selected or rejected, the problem follows the **0/1 Knapsack model**.

Using Dynamic Programming, we constructed a DP table to evaluate all possible cargo selections and determine the optimal subset. The algorithm efficiently calculated the maximum obtainable value while respecting the weight constraint. Backtracking was then used to identify the consignments included in the optimal solution.

This approach is highly suitable for logistics and transportation applications where maximizing profit under limited capacity is an important requirement.

Implementation Details:

1. Created arrays to store cargo weights and payment values.
2. Defined truck capacity as **24 tons**.

3. Constructed a Dynamic Programming table dp[i][w].
4. Applied the 0/1 Knapsack recurrence relation.
5. Computed the maximum payment achievable for each capacity.
6. Retrieved the selected consignments through backtracking.
7. Displayed the optimal cargo subset and total payment value.
8. Analyzed time and space complexity.

Result: Output Screenshots:

```
Maximum Value = ?175 Thousand
Selected Items: [A, C, D, E, H]
Total Weight = 24 tons
Total Value = ?175 Thousand
```

Time Complexity:

- DP Table Construction: **O(n × W)**
- Item Recovery (Backtracking): **O(n)**
- Total Complexity: **O(n × W)**

Where:

- n = Number of cargo items = 8
- W = Truck Capacity = 24

SLA Analysis:

· Total Cargo Items = 8

· Truck Capacity = 24 tons

· Optimal Items Selected = 5

· Total Weight Utilized = 24 tons

· Maximum Payment Earned = ₹175 Thousand

GitHub Link: https://github.com/shashankreddy-21/DSA-Project

Student Signature	**Faculty Signature**